Model Cards

> **Artifacts are not in git.** `dataset/` is gitignored (it was stripped from history), so the `.h5` files these cards describe are not version-controlled and must be provisioned onto any machine that serves them. Both are now rebuildable from source — `scripts/train-skin-cancer-model.py` and `scripts/train-lung-cancer-model.py` — so losing one costs a training run rather than the modality. Back them up anyway: `npm run backup:models`.

Every model this system can serve, what it was measured at, and what it must not be used for. The authoritative machine-readable copy lives in `server/model-availability.ts` and is served at `GET /api/models/cards`.

Reproduce any figure below with:

```
python scripts/evaluate-model.py <model.h5> <data_dir> <class0> <class1>
```

Balanced accuracy (the mean of sensitivity and specificity) is the headline metric, not raw accuracy. On an imbalanced set, a model that answers "negative" for every image scores the majority-class rate while detecting nothing; balanced accuracy scores that degenerate case at 0.5, which is chance.

Lung cancer — ResNet50V2 — ENABLED (retrained 2026-08-18)

Artifact	`dataset/lung_cancer_MRI_dataset/resnet50v2_lung_cancer_model.h5`
Architecture	ResNet50V2 ImageNet trunk, **frozen**; trained Dense(256)+Dropout head
Input	Raw RGB 0–255, 224×224. Normalisation **fused into the graph**
Splits	2575 train / 551 validation / **554 test**, stratified, seed 4242
Evaluation set	554 images (282 cancer / 272 no_cancer), **never used in training or model selection**
Test AUC	**0.88**
Calibration	ECE 0.019 → **0.017** with temperature scaling (T=1.125), **applied**
Deployed operating point	threshold 0.30 on the *calibrated* P(cancer) — *not* argmax
Sensitivity @ 0.30	**0.812** — 229 of 282
Specificity @ 0.30	**0.757**
Balanced accuracy @ 0.30	**0.785**

Reproduce: `python scripts/train-lung-cancer-model.py` then `python scripts/choose-lung-threshold.py`

Why the threshold is not argmax

Argmax implicitly says a missed cancer and a false alarm cost the same. They do not in screening. The full sweep on the held-out set:

THRESHOLD	SENSITIVITY	SPECIFICITY	BALANCED ACC.	MISSED CANCERS	FALSE ALARMS
0.50 (argmax)	0.695	0.982	**0.838**	**86**	5
0.40	0.730	0.941	0.836	76	16
0.35	0.762	0.901	0.832	67	27
0.30 (deployed, calibrated)	**0.812**	**0.757**	0.785	**53**	66
0.25	0.826	0.654	0.740	49	94
0.19	0.911	0.474	0.693	25	143

Argmax has the best balanced accuracy and is the wrong choice: it misses 86 of 282 cancers. The deployed point trades extra false alarms for 33 fewer missed cancers. Override with `LUNG_CANCER_THRESHOLD`.

The threshold is selected on **calibrated** probabilities and applied to calibrated probabilities at inference. Temperature scaling is monotonic so it cannot change the ranking, but it does move where a given numeric cut point sits — selecting 0.28 on raw output and then applying T at inference would silently shift the operating point. The sweep above pre-dates calibration; the deployed row is post-calibration.

Note what the table shows about the model itself: there is no region with both high sensitivity and high specificity. **Roughly 1 in 5 cancers is still missed and 1 in 4 healthy scans is still flagged.**

Why it was retrained rather than just re-measured

The previous figures — 0.75 balanced accuracy, 0.904 sensitivity — came from `lung_cancer_MRI_dataset/validate`, which was the validation generator during that model's own training. The model had been selected against it, so the numbers were optimistic. No untouched data existed anywhere in the repository, so a "held-out test set" could only be created by pooling both directories and re-splitting. The test file list is recorded in `lung_splits.json` so the claim is auditable.

The honest comparison is therefore *not* 0.904 → 0.812. The old sensitivity was measured on data the model had been tuned on and never meant what it said.

Limitations

- Input screening is active: PCA reconstruction error in feature space flags skin images at **100%** and held-out chest images at **0.8%**; pixel checks catch blank and blurred frames, which the feature detector scores as in-distribution.
- Demographic composition of the training data is unrecorded.

- Not clinically validated. Not cleared by any regulator.

This model cannot currently read real clinical imaging

Measured, not inferred. DICOM ingestion was added in August 2026, and the first thing it established is that **this model refuses genuine radiology objects**:

INPUT	OOD SCORE	THRESHOLD	OUTCOME
Held-out `Lung MRI (n).png` from its own training distribution	~10.9	16.51	classified
Real CT DICOM (`pydicom` CT_small, GE RHAPSODE)	**22.88**	16.51	**refused**
Real MR DICOM (`pydicom` MR_small)	**27.16**	16.51	**refused**

Windowing was investigated as a possible cause, and ruled out. A DICOM carrying no VOI LUT was being rendered across its full stored range, which for CT compresses lung parenchyma into the bottom fifth of the greyscale — a genuine defect, since no radiologist reads a chest that way. Correcting it changes nothing about the refusal:

WINDOW APPLIED TO THE SAME CT	OOD SCORE	OUTCOME
Full stored range (the old fallback)	22.88	refused
Lung, WC −600 / WW 1500 (radiologically correct)	**25.04**	**refused**
Mediastinal, WC 40 / WW 400	30.35	refused
Bone, WC 500 / WW 2000	20.32	refused

The correct window is *worse* than the wrong one. There is no presentation of a real acquisition that this model accepts, and tuning the window until one passed would be fitting the preprocessing to defeat the safety check rather than fixing anything. The windowing was corrected because it was wrong, not because it helped.

The refusal is the out-of-distribution detector working correctly — it is doing precisely what it exists to do, and the alternative (a confident verdict on an image type the model has never seen) would be far worse.

But read what it means. The training set is small RGBA PNGs named `Lung MRI (n).png`, of unrecorded provenance, at inconsistent dimensions — consistent with web-scraped or screenshot material rather than DICOM exports. A properly windowed clinical acquisition does not resemble them. So:

> **The lung modality cannot today be pointed at a hospital PACS.** The pipeline around it works end to end — ingest, de-identify, window, screen, classify — and the model at the end of it declines every real object it is shown.

This is the concrete form of the provenance concern already recorded above, and it is the reason retraining on a documented CT dataset with patient-level splits (LIDC-IDRI, NLST) is the first item of clinical work, not a later refinement.

The inference service therefore refuses DICOM for this modality **explicitly**, ahead of the out-of-distribution screen, with a message naming the reason. The screen would catch it anyway, but its wording — "does not resemble the chest images the model was trained on" — reads as though the submitted image were unusual. It is not: every clinical acquisition is refused, always. A radiology department evaluating this needs to hear "this modality cannot read your scanner's output yet", not "try a different image". Until that happens the lung modality is demonstrable only on data shaped like its own training set, and that should be stated plainly wherever it is shown.

Skin cancer — ResNet50V2 — ENABLED (retrained 2026-08-13)

Artifact	`dataset/data/resnet50v2_skin_cancer_model.h5`
Task	Binary classification, `benign` (index 0) vs `malignant` (index 1)
Architecture	ResNet50V2 ImageNet trunk, **frozen**; trained Dense(256)+Dropout head
Input	Raw RGB 0–255, 224×224. Normalisation is **fused into the graph** as a `Rescaling` layer
Training set	`dataset/dataset/data/train` — 2241 images (1224 benign / 1017 malignant) after a 15% validation holdout, ×3 with flip/rotation augmentation
Evaluation set	`dataset/dataset/data/test` — 360 benign / 300 malignant, **never touched during training or model selection**
Balanced accuracy	**0.864** (chance = 0.50)
Sensitivity @ argmax	0.913 — 274 of 300
Specificity @ argmax	0.814 — 293 of 360
Raw accuracy	0.859 (majority-class baseline 0.545)
Validation balanced accuracy	0.873 — close to test, so not overfit to the split

Reproduce: `python scripts/train-skin-cancer-model.py` then `python scripts/evaluate-model.py dataset/data/resnet50v2_skin_cancer_model.h5 dataset/dataset/data/test benign malignant raw_0_255`

The deployed operating point

The service does not use argmax. It bands the malignant probability, and **these are the numbers that describe what a user actually receives**:

TRUTH	→ "BENIGN" (≤0.30)	→ "UNCERTAIN" (0.30–0.70)	→ "MALIGNANT" (>0.70)
Malignant (300)	**10 (3.3%)**	56	234
Benign (360)	268	57	35

- **3.3% of malignant lesions receive an outright benign result.** This is the number to care about: it is the only outcome that actively reassures someone who has cancer.
- 96.7% of malignant lesions are either flagged or escalated to `uncertain`, and both paths route to a clinician.
- 74.4% of benign lesions are cleared; 17% of all scans land in `uncertain`.
- Banding trades label precision for safety: strict "malignant" sensitivity is 0.78, but outright-miss rate falls to 0.033.

Limitations

- **Performance on darker skin cannot be established from this dataset.** See the section below — this was measured, and the measurement's finding is that the data cannot answer the question.
- Not clinically validated. Not cleared by any regulator.
- The trunk is frozen, so the model relies on generic ImageNet features rather than dermatology-specific ones. Fine-tuning the upper blocks would likely help; it was not attempted here because the training box is CPU-only with ~1 GB free.
- 660 test images is a small evaluation set. The confidence interval on 0.864 is wide — roughly ±0.03.

Performance across skin tones

The dataset carries no Fitzpatrick labels, so skin tone was estimated with **Individual Typology Angle** — the standard proxy in dermatology-AI fairness work — computed from the healthy skin surrounding each lesion:

```
ITA = arctan((L* - 50) / b*)     python scripts/measure-skin-tone-performance.py
```

Usable estimates for **511 of 660** test images; the remaining 149 had no identifiable perilesional skin in frame (dermoscope vignetting, or the lesion filling the field) and are excluded rather than guessed at.

TONE (ITA BIN)	N	MALIGNANT	SENSITIVITY	95% CI	SPECIFICITY	OUTRIGHT BENIGN
Dark	4	4	1.000	0.51–1.00	—	0/4
Brown	18	12	1.000	0.76–1.00	0.667	0/12
Tan	42	31	0.968	0.84–0.99	0.909	1/31
Intermediate	45	21	1.000	0.85–1.00	0.875	0/21
Light	86	34	**0.882**	0.73–0.95	0.942	2/34
Very light	316	128	**0.938**	0.88–0.97	0.835	3/128

The finding is the representation, not the sensitivities.

Only **22 of 511 images (4.3%)** are brown or darker. A bin is treated as reliable only with at least 30 malignant *and* 30 benign images, and on that test **just two bins qualify — Light and Very light. Both are light skin.**

So the honest statement is not "the model performs equally well across skin tones". It is: **this dataset cannot tell you how the model performs on darker skin.** The Dark bin contains four images and no benign controls at all; its sensitivity of 1.000 has a confidence interval running from 0.51 to 1.00, which is another way of saying nothing is known.

Across the two reliable bins sensitivity differs by 0.055, with overlapping intervals — no detectable disparity *among light skin tones*, which is not the question that matters.

Do not read the absence of a measured disparity as evidence of fairness. It is evidence that the evaluation set is 96% light-skinned. Closing this requires data, not modelling: a test set with meaningful representation of Fitzpatrick V–VI, ideally with recorded labels rather than an estimate from pixels.

Caveats on the method: ITA is a proxy, not a Fitzpatrick score. It is affected by lighting, white balance and dermoscopy artefacts, and tanning shifts a light-skinned subject darker. It is adequate to detect a large disparity; it is not a substitute for labelled data.

History

The previous artifact scored **0.50 balanced accuracy — exactly chance** — on this same test set, under all three preprocessing schemes tried (raw 0–255 called every image benign; div255 and `resnet_v2.preprocess_input` flagged nearly everything malignant at specificity 0.06). Its training code no longer existed, so its preprocessing and class order could not be recovered, and it was rebuilt from scratch. The app had been advertising that model at "96% accuracy".

Preprocessing is now fused into the saved graph specifically so that inference cannot silently disagree with training about normalisation again.

Modalities with no model

Breast, colon and prostate are named in the UI but **have no trained classifier**. Requests return HTTP 503 and queue the scan for manual review. Until an artifact exists and passes evaluation, they should be removed from the interface rather than presented as capabilities.

What a 503 means

When automated analysis cannot run, the API returns 503 with no diagnostic content and records the scan with status `pending_manual_review`. This is deliberate:

> **A 503 is not a negative result.** It means no model produced an opinion.

Earlier versions of this system filled that gap with `Math.random()` — including a lung path that defaulted to `no_cancer` at 0.5 confidence whenever the model failed to load, which it always did, because the model path pointed at a directory that did not exist on any deployed machine. Nothing in the response distinguished that fabricated negative from a real one.

Generated from MODEL_CARDS.md by scripts/render-doc.py.

Every figure in this document is reproducible with the commands it contains. HealthAI Assistant · 2026 South Africa MedTech Innovation Challenge.